

2.7.23

EE25BTECH11011 - Navya

Question:

For what value of k ($k > 0$) is the area of the triangle with vertices $(-2, 6)$, $(2k, 4)$ and $(2k + 1, 10)$ equal to 52 sq. units?

Solution:

Vertices of triangle are

$$A \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} -2 \\ 6 \end{pmatrix} \quad (1)$$

$$B \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2k \\ 4 \end{pmatrix} \quad (2)$$

$$C \begin{pmatrix} x_3 \\ y_3 \end{pmatrix} = \begin{pmatrix} 2k + 1 \\ 10 \end{pmatrix} \quad (3)$$

$$\text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \quad (4)$$

$$= \frac{1}{2} \begin{vmatrix} -2 & 6 & 1 \\ 2k & 4 & 1 \\ 2k + 1 & 10 & 1 \end{vmatrix} \quad (5)$$

$$= \frac{1}{2} \left[-2 \begin{vmatrix} 4 & 1 \\ 10 & 1 \end{vmatrix} - 6 \begin{vmatrix} 2k & 1 \\ 2k + 1 & 1 \end{vmatrix} + \begin{vmatrix} 2k & 4 \\ 2k + 1 & 10 \end{vmatrix} \right] \quad (6)$$

$$= \frac{1}{2} [(-2)(4 - 10) - 6(2k - (2k + 1)) + (20k - 8k - 4)] \quad (7)$$

$$= \frac{1}{2} |12k + 14|. \quad (8)$$

Given area of triangle = 52 sq.units,

$$\frac{1}{2} |12k + 14| = 52 \quad (9)$$

$$|12k + 14| = 104. \quad (10)$$

Now consider the two cases from the absolute value,

Case 1:

$$12k + 14 = 104 \quad (11)$$

$$12k = 90 \quad (12)$$

$$k = \frac{90}{12} = \frac{15}{2}. \quad (13)$$

Case 2:

$$12k + 14 = -104 \quad (14)$$

$$12k = -118 \quad (15)$$

$$k = -\frac{118}{12} = -\frac{59}{6}. \quad (16)$$

Because the problem requires $k > 0$, we discard the negative solution.

$$k = \frac{15}{2}$$

